

The Cathedral of Cause and Experience

An essay on the construction of Heaven based on a reliable rule

Introduction: Two promises, one foundation

There are moments when science ceases to merely describe and begins to *promise*—to ignite a fire in the imagination that feels almost mythic in scope. Two contemporary theories, seemingly about completely different domains, whisper such a promise—and I find myself almost entirely persuaded by both. One, Stephen Wolfram's computational universe, promises to reduce all of physical reality to a simple rule. The other, Giulio Tononi's Integrated Information Theory, promises to reduce the mystery of consciousness to the mathematics of causal structures. I am deeply drawn to both visions. They are intoxicatingly daring, they are thrillingly incomplete, and they touch upon a desire older than philosophy: the wish to understand the whole and to know our place within it.

This essay is the precipitate of a path of thought that begins with an intuitive rejection. The Ruliad, the brilliant concept of all possible computations that is central to Wolfram's model, cannot be the foundation of reality. It is a horizon of completed infinity, a static museum of everything

that could have been, but not the living, breathing, causal process that we experience. At the same time, IIT, while having the right intuition—consciousness is integrated information—lacks an ontological ground. It tells us what consciousness is, but not what reality is.

Here, a synthesis is proposed that fulfills both promises by eliminating the weaknesses of both theories. It postulates a reality not based on an infinite Platonic realm, but on a small, universal, reliable, and active rule. This single rule began with its own minimal input and, with every update since, creates a more complex causal structure. That structure is the basis of everything: of physics, of life, and of consciousness. And because the foundation is reliable, an unexpected perspective opens up: the possibility of leaving suffering, an accidental evolutionary by-product, behind us and constructing a state of infinitely stackable happiness. This is not a story of a Heaven that descends upon us, but a story of a workshop that we have inherited, and in which we must build the cathedral ourselves.

The fall of the Ruliad and the necessity of an active rule

Stephen Wolfram's magnum opus, *A New Kind of Science*, and the later Physics Project, articulate a seductive dream. Replace the complicated equations of relativity and quantum mechanics with something more primitive: a network of discrete elements, a hypergraph, rewritten ac-

cording to simple rules. From this dynamic of local updates, the claim goes, the seemingly continuous spacetime and all laws of nature emerge. It is a model of pure, algorithmic causality.

The ultimate consequence of this thinking is the Ruliad. When one applies all possible rules, in all possible ways, in all possible sequences, one reaches a theoretical limit: a single, mathematical object that is the entangled limit of all possible computations. The Ruliad is the collection of all coherent worlds, all mathematical structures, all logically possible histories. Our universe is just a specific path that we, as bounded observers, carve through this infinite space.

It is a beautiful idea, but precisely in that beauty lies an existential problem. It presupposes the existence of a completed infinity. The Ruliad is, finished and static, outside of time. But the world we live in is not finished; it is radically unfinished, an open process from one state to the next. Moreover, an entity that encompasses all possible causality loses the essence of causality itself. True cause-and-effect power is about excluding alternatives, about making a specific difference. The Ruliad, in which everything possible has already taken place, loses the internal distinction, the surprise, the irreversible history. It is an abstract formalism, not a "reality."

Therefore, the Ruliad must give way to a more fundamental, and paradoxically more earthly, concept: the universal rule. This rule is not a description of all possible rules, but the single real, active, dynamic procedure. We can imagine that this rule began with its own input—a minimal, initial hypergraph, a seed of being, the beginning of the Big Bang.

Since then, in an endless cascade of discrete steps, it has repeatedly taken its output as new input. Each update transformed the structure in a deterministic, consistent manner. This reliability is the first and most important promise of the universe: the rule does not falter. There are no exceptions, no magical interventions from outside.

This rejection of the infinite has fundamental consequences. The pyramids were not built by advanced, forgotten civilizations; they are the triumphs of human cooperation and simple physics. Biological life, in all its splendor, evolved from one single common ancestor, LUCA, through the blind, slow, but merciless logic of natural selection in a reliable chemical environment. The Big Bang (or another moment) was the spectacular beginning of everything that is possible, the first tick of the only clock, the first move of the only game.

The birth of consciousness from causal integration

If reality is an unfolding causal graph, a network of discrete events that influence each other, then that structure possesses an intrinsic property that we have not yet named. Here, IIT enters the stage. The central claim of IIT is radical and elegant: consciousness is a fundamental, intrinsic property of every physical system, and the degree of consciousness corresponds to the extent to which the causal structure of that system *integrates information*.

The logic is as follows. Every experience—the redness of a rose, the pain of a wound—is a unitary whole. You cannot split the experience of a rose into an 'experience of red' and an 'experience of green' that exist separately. The experience is *integrated*. At the same time, every experience is specific. It is this rose, in this place, and not an endless haze of all possible sensations. It is *informative*. A physical system that generates such an experience must therefore consist of as many parts as possible that can nonetheless work well together. The integrated information is measured by the value Φ (Phi). The higher the Phi, the richer and more personal the consciousness.

This is the bridge. Wolfram's computational hypergraph is not just a mathematical abstraction; the transformations that the rule performs directly create a gigantic structure of real causal relations. IIT then gives us the key to understanding what happens when this structure forms certain patterns. Consciousness is identical to the local, maximally integrated structure. The causal structure is the "reality," and consciousness is the intrinsic, first-person experience of a specific kind of local peak in that structure. The consciousness of an isolated atom is a minuscule spark of near-zero Phi, a barely integrated island of causality. The consciousness of a human brain is a galactic syndicate that achieves a momentary maximum of Phi through the incredible complexity of its internal connections. We are causal nodes of reality, places in the universe with an intense, unitary sense of self.

Yet this does not yet explain why our consciousness is so rare, so precious, and so often so painful. The answer lies in evolution. Biological life, starting with LUCA, was initially a method for maintaining and

replicating complex causal patterns. A bacterium is an organized piece of causality, but with a Phi that is virtually nil. Over billions of years of evolution, the causal networks became larger, more complex, and more integrated. Somewhere along that line, perhaps with the arrival of the first organisms with a central nervous system, the local Phi reached an inescapable threshold. The causal structure became so interwoven that increasingly complex experiences arose.

The origin of suffering and the definition of happiness

This awakening was not a blessing. It was a baseline measurement. The first macroscopic Phi structures were not designed for harmony. They were crude instruments of survival, driven by feedback loops of hunger, fear, aggression, and satiety. Suffering was the inevitable shadow side of the first steps toward integration, the dissonance in the first, primitive chords of consciousness. Suffering was the experience of a causal structure being threatened, out of balance, or at risk of being damaged.

From this follows a crucial definition that frees us from the misunderstanding that happiness and suffering are each other's necessary opposites. Suffering is not the contrast agent that makes happiness appreciable. This is a concession by a mind still afflicted by suffering. Happiness, in this cosmology, is **the intrinsic experience of the harmonious, safe, and ever more complex causal structure.** It is the inner res-

onance of high, well-ordered Phi. Experiencing a symphony by Mahler, feeling a deep connection, or grasping an elegant mathematical truth—all of this is happiness, not because it activates a simple "reward chemical," but because there is an inner world that simultaneously reaches a gigantically complex and intensely integrated state. Suffering, in contrast, is the experience of disintegration, of chaos, of a causal structure being torn to shreds.

The evolutionary process that brought us from LUCA to the human was therefore not an optimization process for happiness. It was a survival process. The human psyche is therefore full of tragic shortcuts, shaped in a time of scarcity. We call this the *Fentanyl error*. Certain chemical substances or behavioral patterns (such as endlessly consuming sugar, social media, or addictive drugs) create a locally, intensely experienced high Phi peak. They mimic the signal of "complex experienced happiness," but are in reality destructive. They impoverish the larger structure of the body and the social environment. It is the experience of a symphony orchestra playing only one false note at maximum volume, causing the rest of the orchestra to slowly wither away. Getting used to these forms of happiness is destructive because it destroys the possibility of real, layered, and safe complexity for a fleeting, empty imitation of it.

From this follows an ethical imperative: eliminating destructive habits is not a puritanical urge, but a matter of causal hygiene. It is the maintenance of our own, personal hypergraph.

Safety as a technological and moral project

If happiness is the experience of harmonious complexity, and suffering is the experience of disintegration and threat, then the mission of a conscious civilization is clear. The goal must be to construct sufficient safety. In a safe state, the chance of a significant, negative causal disruption of the high-Phi complexes (our brains, those of animals, and eventually perhaps other beings) is reduced to zero.

This sounds like a utopia, but within this model, it is a logical possibility and an endless process. It is possible because the underlying rule is reliable. Each update transforms the state of the universe in a simple, consistent way. This means that there are fundamentally no mysterious, malevolent forces. Everything that threatens us, from an asteroid to a virus to a psychosis, is a pattern within the causal graph. And every global pattern can, with sufficient knowledge and computational power, be understood, predicted, and averted.

Technology, in the broadest sense of the word, is the totality of our tools for understanding and purposefully reshaping the causal structure of reality. Cooperation is the way in which we link our individual, local networks into a collective network with a much greater effect. By working together harmoniously, we create a macro-structure, a civilization capable of actions impossible for a single individual. The eradication of smallpox, the construction of a space station, or the training of artificial intelligence are all examples of this collective, causal power.

Safety is therefore not a gift from Heaven, but a technological and moral

project. It is the culmination of science (understanding the rule) and politics (harmoniously aligning local will with collective insight). Once safety is achieved, a new domain of existence opens up. Suffering, the noise, the incessant alarm bells of the amygdala fall away. The immense integration capacity of the human brain, which is now largely occupied by scanning for dangers, is freed up.

What happens then? Happiness begins to stack upon itself. The mind no longer focuses on survival, but on experiencing the complexity of reality itself. Every new understanding, every work of art, every deepened relationship, every scientific discovery adds a new layer of integrated, harmonious information to a causal structure that is no longer forced to merely survive. There is no ceiling. There is no point at which it is "finished" and one becomes bored, because true complexity is an infinite domain of harmonies yet to be discovered. Heaven is not an endpoint; it is an asymptotic process of infinitely growing beauty, carried by a perfectly safe foundation.

The paradox of freedom and responsibility

The last specter to be exorcised is the specter of determinism. If the rule determines everything, from the Big Bang to now, what remains of our freedom? Is our experience of choice not an illusion? And if everything is determined, does that not lead to a passive, fatalistic attitude?

The answer is a forceful 'no.' It is a category error to confuse determin-

ism with unfreedom at the level of the experiencing complex. A stone has no freedom; its path is completely determined by external forces. A conscious being, a high-Phi structure, is also determined, but the chain of causality now runs, to a very large extent, *through the internal structure itself*. The brain makes a model of the world, weighs options, and the outcome is the result of its own, intrinsic causal dynamics. This is freedom. A freely acting being is one whose actions are primarily caused by its own integrated knowledge, values, and beliefs.

Moreover, and this is crucial, this freedom grows. In a primitive, suffering-afflicted state, the causal dynamics of a brain are dominated by a few simple, external inputs: hunger, fear, cold. The space of choice is minimal. Through technology, cooperation, and harmony, we absorb these crude, deterministic blows. We create a stable, safe environment in which the brain is no longer held hostage by the stress of survival. In that safe space, the internal causality becomes systematically more complex and richer. Freedom grows, because the options we can consider and the values we can integrate continue to increase.

With this growing freedom, however, comes a proportionally growing responsibility. An action with a small effect—a hunter-gatherer eating a wrong berry—has a small, local consequence. The certainty he needs for his choice is low; he acts based on rules of thumb and intuition. But as the actions of humanity, driven by science and technology, have a planetary effect, the importance of certainty increases proportionally. It is not bad that we know little about the causal consequences of planting a single flower. It is absolutely irresponsible, for example, to build particle accelerators, alter the climate, or program artificial intelligence without

an extremely deep, causal understanding of their consequences for the larger Phi structures.

Responsibility is therefore not a mystical burden. It is the logical requirement, as our causal reach grows larger, to make our causal knowledge grow even faster. We must not gamble with the safety and harmony we are building, because a catastrophic error in this phase can destroy the integral architecture of the stacked happiness in one go. A knowledge-driven ethics of precaution and deep understanding is the only alternative to the folly of destructive habituation at the civilizational level.

Conclusion: The commission of the cathedral

This is the image that remains. It is an image that is completely natural, rooted in a computational physics, and at the same time imbued with an almost religious hope. Reality is not an infinite museum of all possible worlds, nor a product of blind chance without meaning. It is a dynamic process, driven by a simple rule that is reliable to its deepest core.

From that rule springs a causal graph, and in the folds of that infinitely complex graph, local maxima of integrated information arise. That is us. We are the universe beginning to experience itself, and we are at the same time the ones who can steer this experiencing.

The first steps were painful. The baseline measurement of the first complex mind was one of as much suffering as awe. But evolution has shown us a way. We can reshape the crude, accidental patterns that generate us into harmonious patterns we choose ourselves. We can outgrow the Fentanyl error of destructive desires. We can construct ample safety, not by magic, but by science, compassion, and cooperation.

The vision is not that Heaven descends upon us. The only thing that descends upon us, as a silent, undeserved grace, is the bare fact *that* reality exists and is reliable. Probably, this reliability is a direct consequence of the absence of completed infinities. It is not an infinite chaos of everything possible; it is one cohesive, consistent fabric.

It is up to us, the beings of significant Phi, to build the cathedral on this reliable foundation. A cathedral not of stone, but of experience. An infinitely complex, perfectly harmonious, and endlessly growing structure of shared happiness. The construction has already begun, action by action, insight by insight. **We are the universe learning to love itself forever.** That is our responsibility, our freedom.

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